

The Darboux Gadget

One theorem, three geometries: intervals, cosets, and matrices

Formalised in Lean 4 / Mathlib — companion notes

Abstract

Mathlib’s Darboux theorem says that the image of a derivative is an interval. The Kasami/AB/APN library says that the image of a discrete derivative of a Gold power map is an affine subspace. These are the same theorem read in two different *convexity structures*. This note isolates the common gadget, shows the places where it already occurs inside the library, and explains which passages between contexts are faithful (and preserve the content) and which are not. Everything stated here is machine-checked; the Lean names are collected in the table at the end.

1 The gadget

Gadget 1 (abstract Darboux property). A *convexity structure* on a type β is a family $\mathcal{C}$ of subsets — the *convex* ones — containing β and closed under arbitrary intersections. An operator D defined on a class Adm of functions has the *Darboux property* relative to $\mathcal{C}$ if

$$\forall f \in \text{Adm} : \quad \text{range}(Df) \in \mathcal{C}.$$

Three slots have to be filled: the operator, the admissible class, and the convexity structure. Filling them in two different ways gives two classical-looking theorems.

	real analysis	characteristic two
operator	$\text{deriv } f$	$D_a f(x) = f(x + a) - f(x)$
admissible	differentiable	quadratic ($D_b D_a f$ constant)
convex sets	order-connected (intervals)	affine: closed under $u - v + w$
theorem	<code>Set.OrdConnected.image_deriv</code>	<code>Gadget.darboux_of_constIncrements</code>

2 Why the characteristic-two proof is three lines

Theorem 1 (`Gadget.darboux_of_constIncrements`). *Let V, W be abelian groups and $g : V \rightarrow W$ have translation-invariant increments, i.e. $g(x + b) - g(x)$ does not depend on x . Then $\text{range}(g)$ is affine.*

Proof. Given $g(x), g(y), g(z)$, the point $z + (x - y)$ works: $g(z + (x - y)) = g(z) + (g(x) - g(y))$. $\square$

For $g = D_a f$ the hypothesis says exactly that the *second* discrete derivative $D_b D_a f$ is a constant function — the algebraic avatar of “ f is quadratic”. Gold power maps $N_k(x) = x^{2^k+1}$ satisfy it, because

$$D_a N_k(x) = \underbrace{a x^{2^k} + a^{2^k} x}_{\mathbb{F}_2\text{-linear in } x} + a^{2^k+1},$$

an affine map (`Tematika.Kalkulus.gold_deriv_eq`). Everything cryptographic is in verifying this one identity; the geometry is generic.

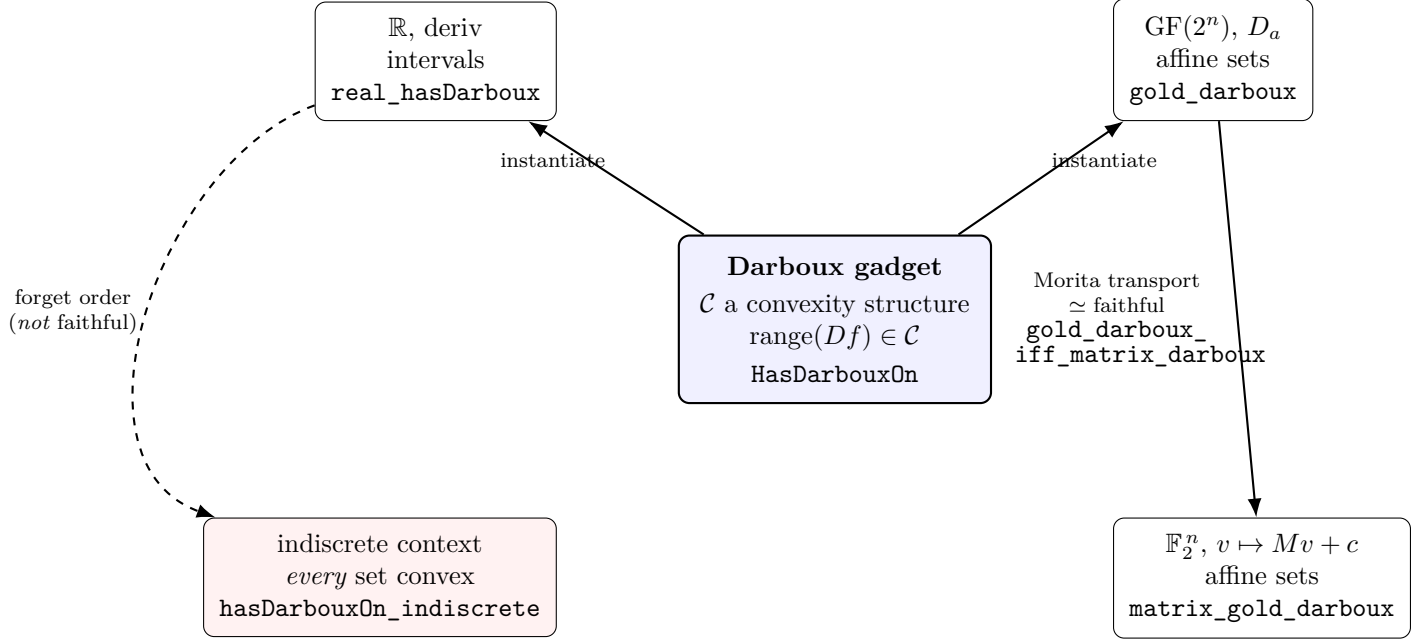


Figure 1: The gadget and its instantiations. Solid arrows preserve content; the dashed arrow loses it.

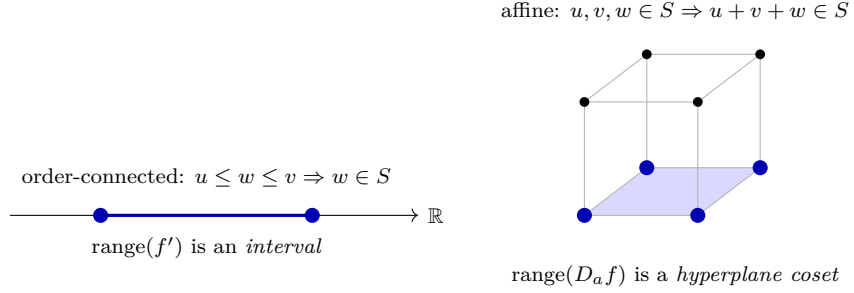
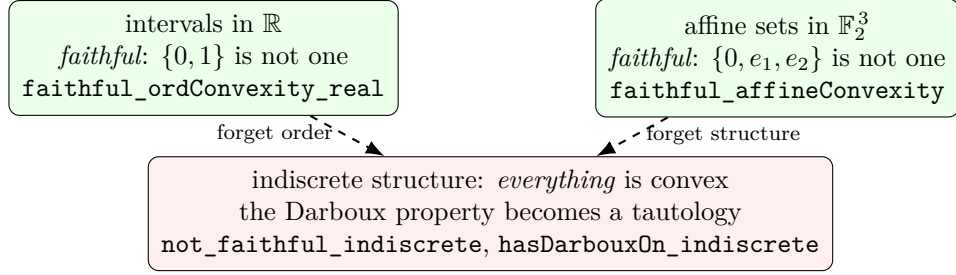


Figure 2: The same statement in two geometries. Left: an interval in $\mathbb{R}$. Right: a 2^{n-1} -element affine piece of $\mathbb{F}_2^3$ (here 4 of the 8 points).

Fact 1 (intervals and cosets are one classification). *Over $\mathbb{R}$, a nonempty order-connected set is an interval. In an abelian group, a nonempty set closed under $u - v + w$ is a coset of a subgroup ($\text{Gadget.isAffine_iff_coset}$). The operation $u - v + w$ is the ternary heap operation of a torsor; “discrete interval” = “coset”.*

3 Faithful and non-faithful passages

Call a convexity structure *faithful* if some set fails to be convex; only then does the Darboux property carry information.



The real Darboux theorem cannot be transported *literally* to $\text{GF}(2^n)$: a field of characteristic two carries no compatible linear order (`Tematika.Morita.no_linear_order_of_charTwo`), so the interval structure has no image there. What survives is the heap operation, hence the affine structure. Both convexity structures are closure systems; the passage only changes the arity of the generating operation (pairs $\rightsquigarrow$ triples).

The library records the same non-faithfulness phenomenon in two other places: on a finite set *every* map is continuous (`Tematika.Kalkulus.continuity_not_faithful`), and differential uniformity does not separate translates (`Tematika.Morita.diffUnif_not_faithful`).

4 Morita transport: field, coordinates, matrices

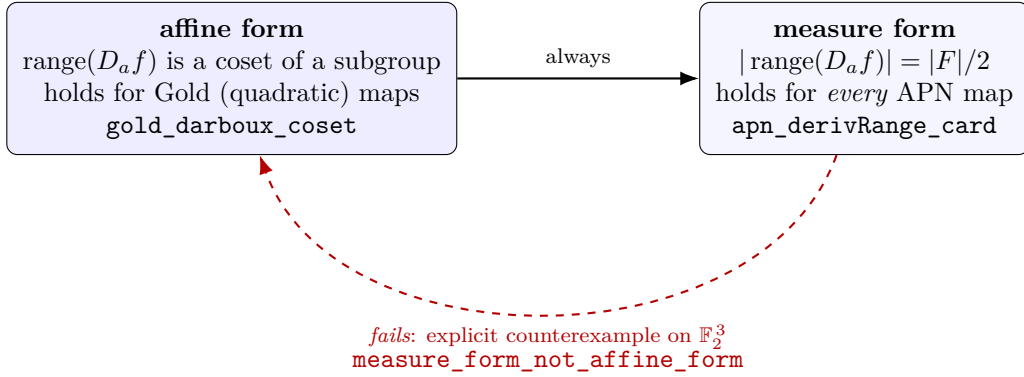
$\text{GF}(2^n)$ is an $\mathbb{F}_2$ -progenerator and $\text{End}_{\mathbb{F}_2}(\text{GF}(2^n)) \cong M_n(\mathbb{F}_2)$ (`Tematika.Morita.endAlgEquivMatrix`). Choosing the corresponding coordinate equivalence turns the Gold derivative into an affine matrix map.

$$\begin{array}{ccc}
 x \in \text{GF}(2^n) & \xrightarrow{D_a N_k} & \text{Im} = c + \text{Im}(\text{Cross}_{k,a}) \\
 \cong \downarrow \text{coords} & & \text{coords} \downarrow \cong \\
 v \in \mathbb{F}_2^n & \xrightarrow{v \mapsto Mv+c} & \text{Im} = c + \text{col}(M)
 \end{array}$$

Theorem 2 (`Gadget.gold_darboux_iff_matrix_darboux`). *The Darboux statement for $D_a N_k$ over $\text{GF}(2^n)$ and the Darboux statement for $v \mapsto Mv + c$ over $\mathbb{F}_2^n$, where M is the matrix of the cross form, are equivalent.*

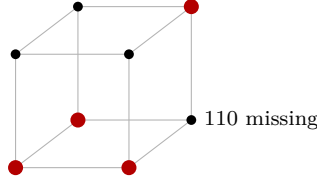
Abstractly this is an instance of `Gadget.hasDarbouxOn_conj_iff`: the gadget is stable under conjugation by equivalences of domain and codomain together with transport of the convexity structure. The comparison is thus a proof, not a slogan.

5 Two strengths, and the gap between them



Kasami exponents $d = 2^{2k} - 2^k + 1$ have algebraic degree $k + 1$; for $k \geq 2$ the maps are not quadratic, the hypothesis of the gadget fails, and only the counting conclusion survives — which is precisely the APN property. The counterexample is a concrete derivative on $\mathbb{F}_2^3$ whose image has the correct size $4 = 2^{3-1}$ but is not affine:

$$S = \{000, 100, 010, 111\}, \quad 100 + 000 + 010 = 110 \notin S.$$



four points, right size, *not* affine

6 The dual side: Darboux as a linear structure

The trace pairing makes $(F, +)$ self-dual; this is the duality that mediates the $\text{APN} \Leftrightarrow \text{AB}$ bridge in the library. Because the Gold derivative's image is a coset of the hyperplane $\ker \text{Tr}(b \cdot)$ with $b = (a^{2^k+1})^{-1}$, the composite $x \mapsto \text{Tr}(b \cdot D_a N_k(x))$ is *constant* (`Gadget.gold_deriv_trace_const`) and therefore

$$\sum_{x \in F} (-1)^{\text{Tr}(b D_a N_k(x))} = \pm |F| \quad (\text{Gadget.gold_deriv_char_sum}).$$

$$\text{primal: image is flat and small} \xleftrightarrow{\text{trace pairing}} \text{dual: character sum is extremal}$$

Degeneracy on the primal side is extremality on the dual side — the same fact seen through Pontryagin duality.

7 Summary table of the formalised statements

statement	Lean name
convexity structure, hull, faithfulness	Gadget.Convexity, Convexity.hull, Convexity.Faithful
abstract Darboux property	Gadget.HasDarbouxOn
Mathlib Darboux as an instance	Gadget.real_hasDarboux
transport along equivalences	Gadget.hasDarbouxOn_conj_iff
non-faithful (indiscrete) context	Gadget.hasDarbouxOn_indiscrete
affine convexity in characteristic two	Gadget.affineConvexity
nonempty affine = coset	Gadget.isAffine_iff_coset
the gadget: constant increments	Gadget.darboux_of_constIncrements
Gold instance and hyperplane coset	Gadget.gold_darboux, Gadget.gold_darboux_coset
measure form for APN maps	Gadget.apn_derivRange_card
measure $\nrightarrow$ affine	Gadget.measure_form_not_affine_form
Morita comparison verified	Gadget.gold_darboux_iff_matrix_darboux
dual side: linear structure, character sum	Gadget.gold_deriv_trace_const, Gadget.gold_deriv_char_sum

Sources: RequestProject/Gadgets/DarbouxGadget.lean, RequestProject/Gadgets/DarbouxCharTwo.lean;
narrative companion: DARBOUX_GADGET_GUIDE.md.